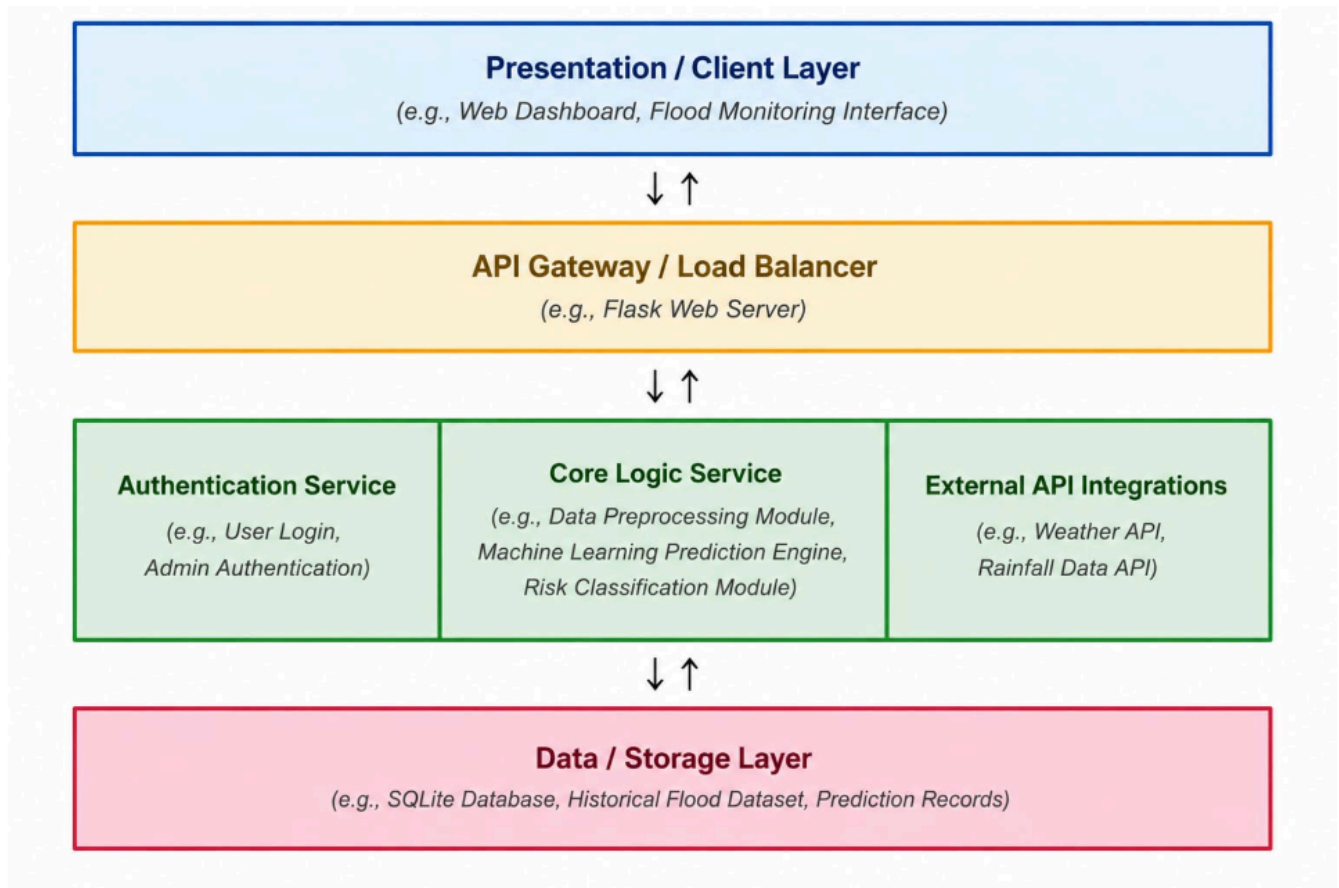


Solution Architecture

Date	15 July 2026
Team ID	SWTID-2026-3798
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

The system collects flood-related data, processes it using machine learning, and displays flood predictions through a web dashboard. The predicted results are stored in a database and shown to users as alerts.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides user interface for monitoring flood risks and alerts</i>	<i>HTML, CSS, Flask</i>
<i>API Gateway</i>	<i>Handles incoming requests and routes them to services</i>	<i>Flask</i>
<i>Authentication Service</i>	<i>Manages user and administrator login</i>	<i>Flask Authentication</i>
<i>Core Logic Service</i>	<i>Processes data and performs flood prediction</i>	<i>Python, Scikit-learn</i>
<i>External APIs</i>	<i>Retrieves weather and rainfall information</i>	<i>Weather APIs</i>
<i>Database Layer</i>	<i>Stores historical flood data and prediction results</i>	<i>SQLite</i>